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Abstract

EXAMINATION OF THE EFFECT OF ILLEGAL PARKING ON CAPACITY REDUCTION OF URBAN ROADS (CSCE’24)

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ABSTRACT

The city of Karachi has more than 20 million residents who face severe traffic congestion due to infrastructure problems and inefficient traffic management. One of the reasons for the congestion is the habit of illegal parking, which results in the formation of bottlenecks. This study investigates the impact of illegal parking on traffic congestion, which is a major cause of air pollution and increased emission levels. On-site surveys were conducted to measure vehicle flow, illegal parking extent, and operational lanes during peak hours. Data were collected manually at Allama Shabbir Ahmed Usmani Road, in Gulshan-e-Iqbal Town. It was found that an average of 2.05 lanes are operational, catering to a traffic volume of 2,575 PC/hr in front of Blue Ribbon Bakery, while 1.88 lanes are operational with a traffic volume of 3,144 PC/hr in front of Manpasand Foods. The calculated road capacity was found to be 2,075.6 PC/hr/lane. This slightly higher peak-hour volume indicates a near-saturation condition, and congestion is inevitable. This congestion leads to increased fuel consumption and emissions, adversely affecting commuters' health. The study underscores the need for improved traffic management to mitigate these issues.